

EDUCATION

University of Toledo | B.S. Mathematics (Statistics Concentration) with Minor in Data Science, B.A. in Economics | GPA: 3.88 | Dean's List (Freshmen year)

Coursework: CSET-I, Intro to Statistics, Calculus I & II, Statistical Methods I (InProgress), Elementary Linear Algebra (InProgress), Multivariable Calculus (InProgress), MATH 4900 — Senior Seminar, AI Algorithms in Chess (InProgress)

SKILLS

- **Technical Skills:** Python, PostgreSQL, Scikit-learn, SQLite
- **Tools & Platforms:** Git/GitHub, VS code, Jupyter Notebook, Claude, Cursor, Microsoft Excel

PROJECTS

SnaiCell | Python, XGBoost, Scikit-learn, pandas — ML pipeline (RocketHacks 2026) March 2026

- Engineered 28 domain-derived features from raw stroke imaging and hemodynamic data (penumbra salvageable ratio, cerebral oxygen reserve, hemorrhagic risk, clot burden, brain deterioration rate)
- **Trained XGBoost bad-outcome classifier on 3,000 patient records with stratified split and scale_pos_weight class-imbalance correction; achieved 0.82 ROC-AUC on a held-out 30% split, with mutual-information selection cutting 84 features to 30 at no loss in discrimination**
- Integrated Featherless-hosted OpenBioLLM-70B for real-time AI emergency triage — natural language patient descriptions automatically mapped to full pre-operative clinical profiles

Neural Network from Scratch | Python, NumPy | Feb 2026 – April 2026

- Implemented neurons, dense layers, and batch processing from Math and Numpy
- Implemented forward propagation, ReLU activation, Softmax function, and dense layers from scratch
- Implemented Backpropagation, updating parameters to get 90% accuracy on MNIST dataset.

Instagram User Analytics Dashboard Dec 2025 – Jan 2026

- Processed 1.2M-row Kaggle dataset, sampling 50K users for fast and accurate analysis.
- Built 8+ interactive visualizations for demographics, engagement, and content trends.
- Analyzed gender, country, relationships, and activity metrics (sessions, reels, stories, likes, DMs).

EXPERIENCE

IT Lab Assistant Dec 2025 – Present
University of Toledo

- Provided technical support resolving hardware and software issues for 50-100 users weekly
- Maintained system uptime and managed IT inventory and hardware
- Assisted students and faculty with software troubleshooting and system optimization

CERTIFICATIONS

- **CS50P: Introduction to Programming with Python — Harvard University**
- **CS50SQL — Harvard University**
- **Machine Learning Specialization — DeepLearning.AI**

INVOLVEMENT | Runner-up, USCF U1000 Canton Chess Tournament | Outstanding People's Choice Award, Fall 2025 Showcase | AI/ML Club Member | UToledo Data Club Member
